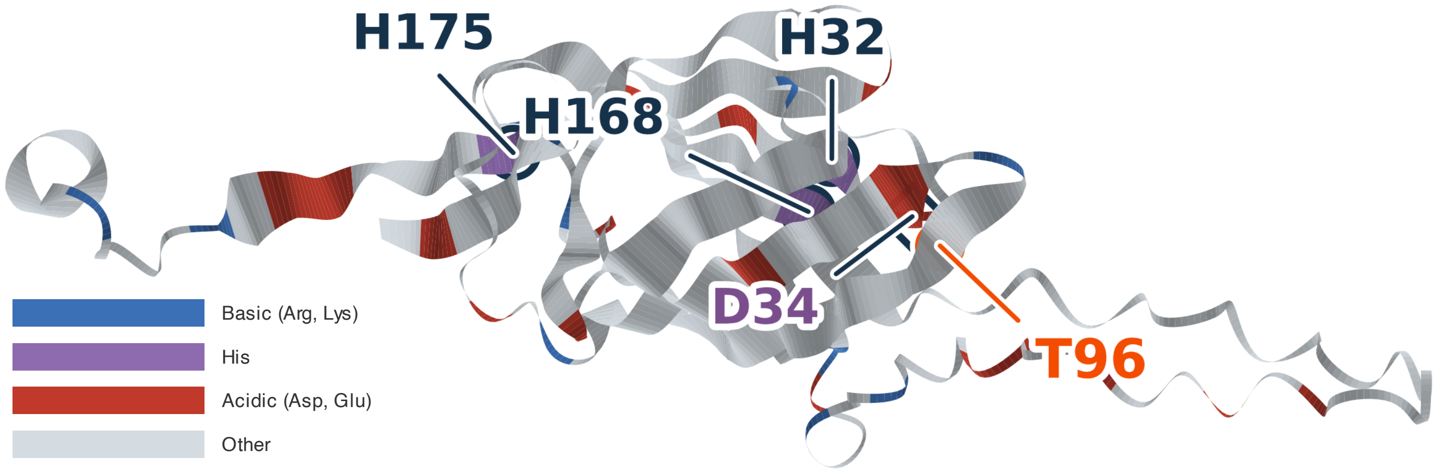
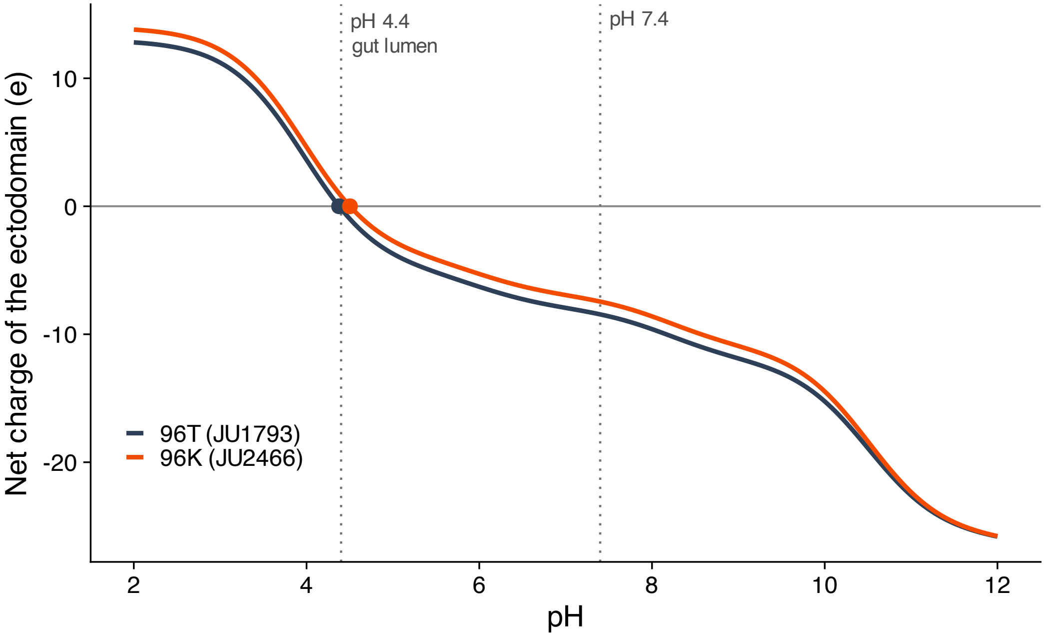
A Charged residues on the ectodomain

AlphaFold3 model, residues 21–193 of chain A, coloured by residue class. Histidine is separated because it is the class that titrates between neutral pH and the acidic gut lumen. T96 in orange; the three uptake-critical histidines of McEwan et al. 2012 labelled in dark blue; D34, the qt13 loss-of-function allele, in purple, because it is a separate line of evidence and not one of the histidines.



B At gut pH the variant changes the sign of the surface

Henderson–Hasselbalch over the side chains of residues 21–193; termini omitted. The ectodomain carries -8.4 e at pH 7.4 but only -0.2 e at the gut-lumen pH of 4.4 — its isoelectric point, 4.38, is essentially the pH it works at (dots). Against that near-neutral background the +1 e from T96K takes the surface from -0.2 e to +0.8 e and the isoelectric point to 4.50. The variant is a small change in absolute terms, but at lumenal pH it is the difference between a slightly negative and a slightly positive face — and 96K is the allele with efficient uptake.



C Close, but not more than chance would give

Distances from residue 96 to every other ectodomain residue, with the three uptake-critical histidines marked (dark blue) and D34, the qt13 allele, shown separately (purple, dashed) because it is a different experiment and is not in the statistic. 2 of the 3 histidines are nearer than the median (23.8 Å), but 41% of the ectodomain lies within 20 Å, so 2 of 3 landing there is not surprising: binomial $p = 0.37$, and a permutation test on their mean distance gives $p = 0.47$. Read this as spatial context for T96, not as evidence of a shared site.